ChartOCR-SourceDiag: Diagnosing Source-Channel Failures in Chart and OCR-Centric Multimodal Reasoning

Anonymous ACL submission

Abstract

Chart question answering and visual-text reasoning benchmarks increasingly stress multimodal systems, but aggregate accuracy often hides whether a model used chart marks, OCR-visible text, or prompt shortcuts. We introduce ChartOCR-SourceDiag, a diagnostic evaluation package that organizes three real source families—ChartQAPro, CharXiv, and OCRBench-v2—into source-channel slices, prompting controls, and scoring-scope labels. In the executed matrix, seven conditions are evaluated over 900 unique tasks and 6,300 scored rows: two hosted image-conditioned backends, LLaVA-1.5-7B, ChartInstruct-FlanT5-XL, question-only, OCR-text-only, and abstain-calibrated visual prompting. Direct visual prompting substantially improves over question-only controls by a paired mean score of 0.5010 and over OCR-text-only controls by 0.5443 across the executed families, while the second hosted image backend is competitive with the original hosted visual condition. The analysis separates automatic scoring on ChartQAPro and OCRBench-v2 from PairScorer-assisted CharXiv validation, and identifies source-channel slices where text-only and smaller open/chart-specialized baselines remain weak. The benchmark therefore supports a scoped diagnostic claim: source-channel controls reveal chart/OCR failure patterns that a single aggregate score would obscure, while keeping model identity, answer budget, and scorer scope visible for future comparisons.

1 Introduction

Charts combine visual marks, text, legends, axes, layout, and often implicit data tables. A multimodal system answering chart questions can succeed by reading plotted values, by copying OCR-visible text, by exploiting answer priors, or by using source-specific shortcuts. This makes chart plus OCR robustness a poor fit for a single undifferentiated score. Chart QA benchmarks have moved from controlled synthetic figures toward realistic and diverse chart questions (Kafle et al., 2018; Kahou et al., 2017; Methani et al., 2019; Masry et al., 2022, 2025). OCR and document VQA benchmarks show a parallel pressure point: visual text and layout are core reasoning inputs, not merely preprocessing artifacts (Singh et al., 2019; Biten et al., 2019; Mathew et al., 2020; Liu et al., 2023b; Fu et al., 2024). Scientific-chart resources such as CharXiv further stress whether a model binds visual evidence, text, and figure context under realistic paper-figure conditions (Wang et al., 2024). The gap is not a lack of another aggregate score; it is the lack of a diagnostic view that keeps source channel, control condition, and scorer scope visible when an image-conditioned backend improves.

This paper constructs a diagnostic resource and analysis package for that question. ChartOCR-SourceDiag does not introduce a synthetic task family. Instead, it samples real tasks from three public source families and records the source-channel role of each family: diverse real-world charts, scientific chart figures, and OCR-heavy visual text cases. Its diagnostic insight is to treat evidence channels as controlled sources of variation. If question-only or OCR-text-only rows approach an image-conditioned score, the aggregate score may not demonstrate visual chart grounding; if those controls collapse while image-conditioned rows improve, the evaluation has a concrete source-channel signal. The benchmark then evaluates seven controlled conditions: two hosted image-conditioned backends, one open-weight image-conditioned baseline, one chart-specialized baseline, question-only prompting, OCR-text-only prompting, and an abstain-calibrated visual prompt. The goal is not to claim a broad model ranking. It is to expose where image evidence matters, where text controls fail, which scoring scopes are automatic, and which claims must remain bounded by the executed evidence.

The resulting matrix contains 6,300 scored rows from 900 unique tasks. The central result is simple but useful: direct visual prompting beats question-only prompting by 0.5010 paired mean score and beats OCR-text-only prompting by 0.5443 over the executed families. These gains are large in ChartQAPro and OCRBench-v2 under automatic scoring, and they also appear in CharXiv under LLM-assisted validation. The second hosted image backend is comparable to the original hosted visual condition, whereas LLaVA-1.5-7B and ChartInstruct-FlanT5-XL are much lower in this run. In contrast, abstain calibration is not a uniformly stronger replacement for direct visual prompting: its overall paired delta is 0.0044 and its family-level effects are small and mixed.

The contribution is therefore a scoped benchmark-and-analysis package: (1) a source-channel diagnostic organization over three real benchmark families; (2) an execution matrix with hosted visual, second hosted visual-text, open-weight LLaVA, ChartInstruct, question-only, OCR-text-only, and abstain-calibrated controls; (3) automatic scoring for ChartQAPro and OCRBench-v2, with a separately labeled PairScorer-assisted CharXiv slice and duplicate-grader consistency audit; and (4) result and failure-taxonomy artifacts that bind claims to raw scored rows. We explicitly do not claim exhaustive open-model coverage, a broad chart-specialized or derendering-system comparison, fully automatic CharXiv scoring, OCRBench core-image evidence, or a leaderboard over the full API model ecosystem.

The paper is organized to make those boundaries inspectable. Section 3 defines the source families, channel labels, and prompt controls before showing results. Section 4 names the evaluated hosted model identifiers, scoring scopes, budgets, and paired comparison policy. Section 5 reports the cross-family matrix and paired deltas. Section 6 turns the deltas into a failure-pattern analysis and names the claims that must remain future work. This ordering follows the diagnostic thesis: evidence provenance and scoring scope are part of the benchmark, not an appendix-only detail.

2 Related Work

Chart question answering and visualization reasoning. Early chart and visualization QA resources established the task as a combination of visual reading and symbolic reasoning. DVQA and FigureQA emphasized structured visual reasoning over plots and synthetic figures (Kafle et al., 2018; Kahou et al., 2017); PlotQA added scientific plots and real-valued answers (Methani et al., 2019); ChartQA brought visual and logical reasoning over charts into a modern QA benchmark setting (Masry et al., 2022). Recent resources increase chart diversity and reasoning complexity: ChartBench stresses complex chart reasoning (Xu et al., 2023), ChartQAPro expands source diversity and question types (Masry et al., 2025), ChartMind studies complex real-world multimodal chart QA (Wei et al., 2025), and ChartQA-X connects chart reasoning with explanation generation (Hegde et al., 2025). ChartOCR-SourceDiag uses this line of work as source evidence, but organizes the evaluation around source-channel controls rather than a new aggregate leaderboard.

OCR-centric multimodal evaluation. Document and text-centric VQA benchmarks show that OCR and layout are core visual reasoning skills rather than preprocessing details. TextVQA and ST-VQA foreground reading text in images (Singh et al., 2019; Biten et al., 2019); DocVQA focuses on document images (Mathew et al., 2020); OCRBench and OCRBench-v2 broaden the evaluation of visual text localization and reasoning in multimodal models (Liu et al., 2023b; Fu et al., 2024). Chart reasoning inherits these issues because axis labels, legends, tick values, and annotations are visually embedded text. Our OCR-text-only control is therefore diagnostic: it tests whether transcript-like evidence is enough, but it is not a chart-specialized derendering or programmatic reasoning baseline.

Multimodal models and chart-specialized systems. General multimodal models and visual instruction-tuned systems motivate broad interest in chart/OCR evaluation (Alayrac et al., 2022; Li et al., 2023; Dai et al., 2023; Liu et al., 2023a; Bai et al., 2023; Peng et al., 2023; OpenAI et al., 2023; Anil et al., 2023). Chart-specialized models and methods attempt to bridge perception, table reconstruction, and reasoning, including UniChart, ChartLlama, DePlot, MatCha, VProChart, GoT-CQA, and ChartAgent (Masry et al., 2023; Han et al., 2023; Liu et al., 2022a,b; Huang et al., 2024; Zhang et al., 2024; Kaur et al., 2025). The executed matrix includes ChartInstruct-FlanT5-XL as one chart-specialized image-conditioned baseline, but it does not include derendering systems or a broad chart-specialized model-family sweep. We therefore use the broader chart-specialized literature as context while limiting claims to the executed ChartInstruct condition.

Diagnostic benchmark design. Modern multimodal benchmarks increasingly test targeted failure modes rather than only average accuracy. CharXiv focuses realistic scientific chart understanding (Wang et al., 2024); MIHBench studies multi-image hallucinations (Li et al., 2025); OmniSpatial and HoloCount stress spatial reasoning and counting (Jia et al., 2025; Deng et al., 2026). Model-family work such as LLaVA and mPLUG-Owl variants also shows that instruction tuning and multimodal pretraining choices affect reasoning behavior (Liu et al., 2023a; Ye et al., 2023, 2024). ChartOCR-SourceDiag follows the diagnostic-benchmark direction but centers a narrower question: how source-channel controls change conclusions about chart/OCR robustness.

Scoring reliability and source attribution. The benchmark also borrows a lesson from VQA and chart-to-text evaluation: an answer score is only useful if the evidence channel and scorer are clear. General VQA, visual text QA, and chart-to-table pretraining work show that visually grounded answers can be entangled with language priors, OCR pipelines, and table reconstruction assumptions (Agrawal et al., 2015; Singh et al., 2019; Lee et al., 2022; Liu et al., 2022a,b). ChartOCR-SourceDiag therefore treats question-only and OCR-text-only rows as diagnostic controls rather than weak baselines to be dismissed. It also keeps CharXiv separate from automatic-scored sources because the evaluated CharXiv slice uses LLM-assisted validation, even though its duplicate-grader spot audit is internally consistent.

3 Method: ChartOCR-SourceDiag Protocol

Figure 1 summarizes the paper-facing protocol. The benchmark starts from three source families, then builds a diagnostic matrix around source-channel labels, scoring-scope verification, and prompting conditions. The source families are selected for complementary stress: ChartQAPro covers diverse real-world charts, CharXiv covers scientific-paper chart figures and reasoning questions, and OCRBench-v2 covers OCR-heavy visual text and localization-style reasoning. Each family contributes 300 tasks in the executed split, for 900 unique tasks.

The protocol separates four types of evidence. First, it records benchmark provenance and task split information. Second, it records source-channel labels such as OCR-visible text, visual mark/value extraction, legend binding, table-like numeric recovery, and text-layout reasoning. These descriptive labels may overlap; they are not causal interventions. Third, it applies scoring-scope labels. ChartQAPro and OCRBench-v2 use automatic scoring in this evaluation. CharXiv is labeled as LLM-assisted validation with a duplicate-grader consistency check. Fourth, the matrix evaluates prompting conditions that remove or modify image evidence.

Source-family construction. The three-source design is intentionally heterogeneous. ChartQAPro contributes chart questions where plotted values, text, and table-like numeric recovery are interleaved. CharXiv contributes scientific figures whose questions often depend on legend binding and text-layout reasoning in a paper-figure setting. OCRBench-v2 contributes OCR-heavy image questions where text localization and reading are central. The executed split fixes the same count for each family—300 tasks—so family-level differences are not driven by different sample sizes. This is also why the paper reports per-family metrics before any combined summary.

Channel labels. The channel labels are attached as analysis fields, not as exclusive dataset classes. A ChartQAPro item can involve both visual mark/value extraction and OCR-visible text; a CharXiv item can involve legend binding and layout; an OCRBench-v2 item can require text reading and layout. The labels are used to summarize which controls fail in which evidence context. They should not be read as a manual causal explanation for every individual wrong answer. The failure taxonomy later operationalizes a failure as a score shortfall under the row scorer, preserving the distinction between measured score behavior and human semantic diagnosis.

The seven conditions are controls and scoped baselines, not a full product leaderboard. Direct visual prompting receives the image and question through the primary hosted visual backend. A second hosted visual-text backend receives the same image-conditioned task. LLaVA-1.5-7B

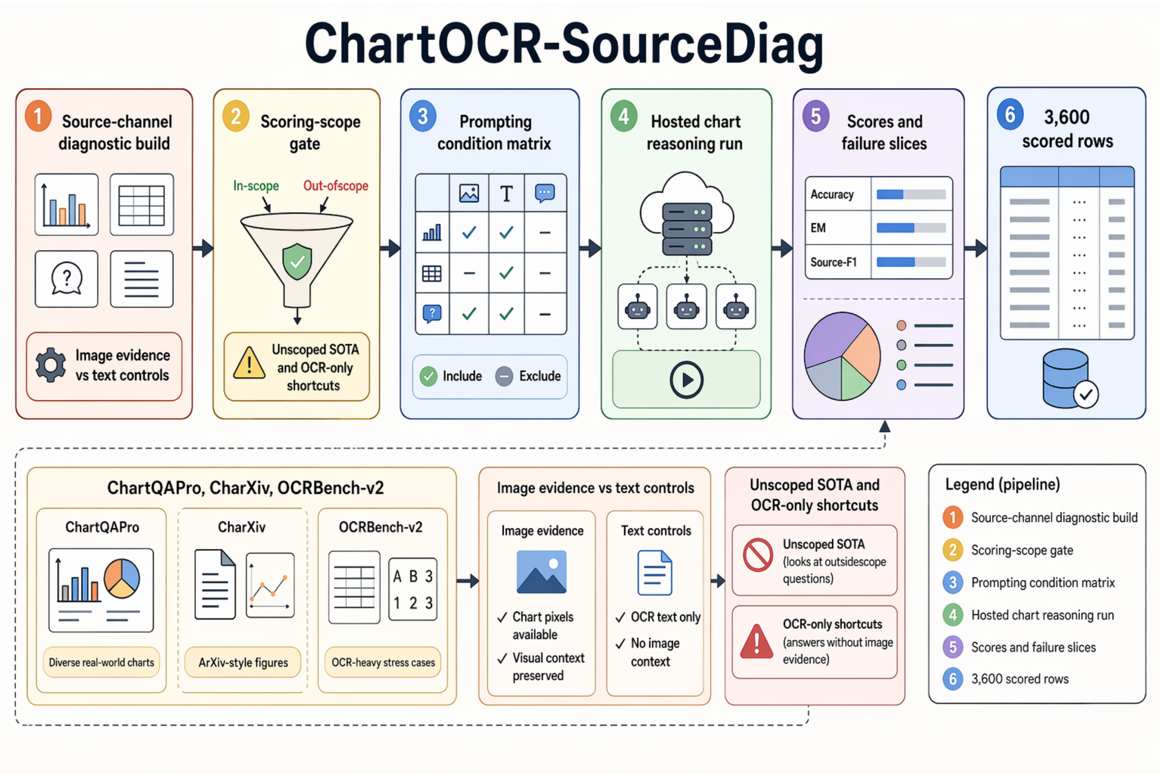

Figure 1: ChartOCR-SourceDiag converts three real chart/OCR source families into a scoped diagnostic matrix over source channels, scoring scope, and prompting conditions. The executed evidence contains 6,300 scored rows across seven conditions and keeps automatic ChartQAPro/OCRBench-v2 results separate from LLM-assisted CharXiv validation.

Source family	Role in the diagnostic protocol	Scoring scope
ChartQAPro	Diverse real-world charts; 300 test tasks; relaxed numeric/ANLS accuracy.	Automatic
CharXiv	Scientific-paper chart figures; 300 validation tasks; duplicate-grader audit for the scored slice.	LLM-assisted
OCRBench-v2	OCR-heavy visual-text cases; 300 test tasks; official VQA accuracy.	Automatic

Table 1: Three source families provide 900 unique tasks; ChartQAPro/OCRBench-v2 are automatic-scored and CharXiv is LLM-assisted.

supplies an open-weight image-conditioned baseline. ChartInstruct-FlanT5-XL supplies a chart-specialized image-conditioned baseline. Question-only prompting removes visual evidence and tests prior or language-only shortcuts. OCR-text-only prompting supplies transcript/control text when available and tests whether text extraction alone is sufficient. Abstain-calibrated prompting keeps the image but asks the model to abstain when evidence is insufficient. Because the abstain condition is a prompt intervention rather than a trained calibration model, we interpret it as a diagnostic ablation.

What the controls isolate. Question-only rows estimate how much can be recovered from language priors, answer-format regularities, or benchmark artifacts. OCR-text-only rows estimate whether extracted or supplied text is enough when the original layout and visual marks are removed. Direct visual rows give the image-conditioned reference condition for the hosted backend. Abstain-calibrated rows test a common prompt-level repair: telling the model to withhold an answer when evidence is uncertain. The design is deliberately conservative. If a text-only control performs well, the corresponding source family may not require much visual grounding for the sampled questions. If it performs poorly while direct visual prompting improves, the benchmark has evidence that image-conditioned access matters for that slice.

4 Experimental Setup

The executed matrix contains 6,300 scored rows: three source families, 300 tasks per family, and seven conditions per task. The paper-facing hosted identifiers are `gpt-5` visual for direct and abstain rows, `gpt-5` multimodal-text for the second hosted image-conditioned row, and `gpt-5` text for question-only and OCR-text-only rows. Hosted calls are single-output requests with no reasoning expansion, low-verbosity answers, no temperature or top-p override, and fixed answer caps: 96 tokens for image-conditioned rows and 64 for text-only rows. The evaluated open-weight and chart-specialized baselines are LLaVA-1.5-7B with greedy 64-token generation and ChartInstruct-FlanT5-XL with deterministic four-beam 96-token generation. Each row records the request modality, evaluated backend, scorer, score, and source-channel labels.

Backend and budget reporting. The benchmark driver is a single-turn source-channel evaluation harness: it builds one prompt from the condition inputs, obtains one short answer, and applies the family scorer to that answer and source-family target. Provider routing, latency, and cost are outside the scientific comparison and are not used as method features.

Metrics are source-specific. ChartQAPro uses relaxed numeric/ANLS accuracy; OCRBench-v2 uses official VQA accuracy; CharXiv uses PairScorer-assisted validation accuracy. For each CharXiv row, PairScorer receives the question, gold validation answer, and model answer, then returns strict JSON with an extracted answer and a binary score; invalid JSON is an evaluation error, not a coerced zero. A 30-row duplicate-grader audit records 30/30 agreement, so CharXiv is reported as LLM-assisted evidence rather than as a fully automatic metric. For comparisons, we use same-task paired deltas and bootstrap confidence intervals. The combined "all executed families" row is descriptive because it mixes scoring scopes; family-level rows preserve the scope label. The failure taxonomy uses source-channel labels, scores, OCR-text availability, and condition comparisons, not manual semantic annotation.

Pairing and uncertainty. Every paired comparison uses the same task under two prompting conditions, so the paired delta asks whether changing only the source channel or abstain instruction changes the row score. Bootstrap intervals are reported for the paired mean deltas. The table also records a sign-flip probability, but the paper avoids significance language beyond whether the confidence interval excludes zero. This matters for abstain calibration: its overall delta is small and its interval crosses zero, so the correct interpretation is mixed behavior rather than a reliable improvement.

The strongest remaining unexecuted comparisons are documented rather than hidden. The matrix includes one open image-conditioned MLLM baseline, one additional hosted visual backend, and one chart-specialized model; broader open-model sweeps, multiple chart-specialized families, derendering systems, OCRBench core-image evidence, and fully automatic CharXiv scoring remain outside scope.

Matrix completeness. All 21 family-condition cells contain 300 scored tasks. Invalid, errored, or empty model outputs are evaluated under the corresponding family scorer instead of being replaced, so the matrix is complete for the scoped seven-condition analysis while broader model-family and derendering conditions remain limitations.

5 Main Results

Table 2 is the central matrix. It shows a consistent pattern: hosted image-conditioned backends are far above question-only and OCR-text-only controls in every source family, while LLaVA-1.5-7B and ChartInstruct-FlanT5-XL are much lower in this run. The absolute scores differ by source and metric, so the table should be read within each family rather than as a single cross-source leaderboard. ChartQAPro is the hardest automatic-scored slice, OCRBench-v2 has the highest hosted visual score, and CharXiv sits between them under LLM-assisted validation.

Figure 2 plots the same family-condition pattern. It is useful because the controls and smaller open-weight/chart-specialized baselines fail in a visually similar way across sources even though the source metrics differ. ChartQAPro hosted visual scoring remains modest at 0.3828 for the strongest hosted condition, so the benchmark should not be read as solved. CharXiv reaches 0.6833 under the second hosted visual backend, but the scorer is LLM-assisted. OCRBench-v2 direct visual reaches 0.8367, but that is the OCRBench-v2 source rather than OCRBench core. These qualifications are not side notes; they are exactly the reason the matrix includes condition, backend, budget, and scoring-scope fields.

Table 3 reports the paired deltas. The overall direct-visual gain over question-only is 0.5010 with

Condition	Evaluated model/backend	Role	Budget / decoding	ChartQAPro auto, 300 test	CharXiv PairScorer, 300 val.	OCRBench-v2 auto, 300 test
Direct visual	`gpt-5` visual backend	Image reference	Image + question; 96-token cap	0.3560	0.6233	**0.8367**
Hosted V-T	`gpt-5` multimodal-text backend	Second hosted image backend	Image + question; 96-token cap	**0.3828**	**0.6833**	0.8241
LLaVA	LLaVA-1.5-7B	Open-weight baseline	Image + question; 96-token cap	0.0792	0.1133	0.2185
ChartInst.	ChartInstruct-FlanT5-XL	Chart-specialized baseline	Image + question; 96-token cap	0.0191	0.1300	0.2374
Question only	`gpt-5` text backend	Language-only control	Question only; 64-token cap	0.0606	0.1200	0.1324
OCR only	`gpt-5` text backend	Text-channel control	OCR text only; 64-token cap	0.0446	0.0667	0.0718
Abstain	`gpt-5` visual backend	Prompt intervention	Image + abstain instruction; 96-token cap	0.3619	0.6400	0.8274

Table 2: Seven-condition matrix over 6,300 scored rows with backend and budget visible: hosted image-conditioned backends reach 0.3828–0.8367 by source family, while text-only controls stay below 0.1324.

a 95% bootstrap interval of [0.4672, 0.5357]. The gain over OCR-text-only is even larger at 0.5443 with interval [0.5117, 0.5777]. These comparisons support the source-channel thesis: the image is not merely a container for text. Removing it leaves both question-only and OCR-text-only controls far behind the visual condition.

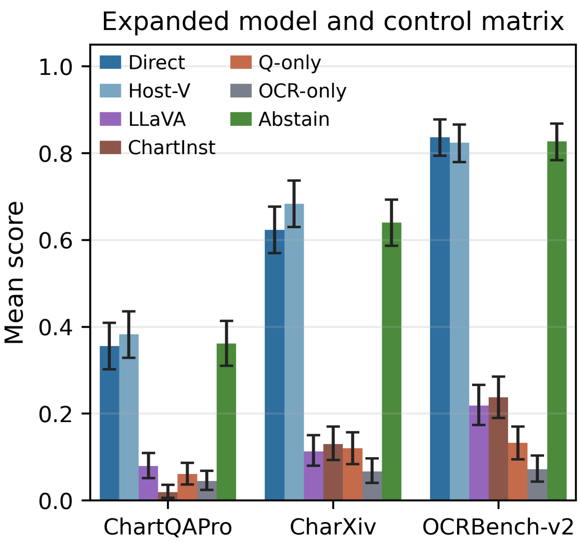

Figure 2: Family-level means mirror Table 2: hosted image-conditioned prompts dominate text-only controls and the open/chart-specialized baselines, while absolute levels differ by source and scoring scope.

The abstain result is deliberately not framed as a win. On ChartQAPro, abstain-calibrated prompting slightly exceeds direct visual prompting; on OCRBench-v2 it is slightly below; and in the overall paired row the interval crosses zero. This matters because abstention language is a common prompt-level safety repair. In this executed matrix it changes behavior, but it does not replace visual evidence or solve the underlying source-channel failures.

The family-level paired rows sharpen this point. Direct visual exceeds question-only by 0.2954 on ChartQAPro, 0.5033 on CharXiv, and 0.7044 on OCRBench-v2. Direct visual exceeds OCR-text-only by 0.3113, 0.5567, and 0.7650 on the same families. All six intervals exclude zero. By contrast, abstain-calibrated minus direct visual is 0.0059 on ChartQAPro, 0.0167 on CharXiv, and -0.0094 on OCRBench-v2, with all three intervals crossing zero. The most defensible conclusion is therefore asymmetric: source-channel controls are strongly diagnostic, while abstain wording is at best a family-dependent prompt variation.

6 Paired Analysis and Failure Patterns

Table 3 is kept with the paired-analysis discussion because it is the compact numerical evidence behind the deltas in Figure 3. The table separates

Paired comparison	Delta	95% interval
Direct visual – question only	**0.5010**	[0.4672, 0.5357]
Direct visual – OCR text only	**0.5443**	[0.5117, 0.5777]
Abstain calibrated – direct visual	0.0044	[-0.0137, 0.0224]

Table 3: Across 900 paired tasks, visual evidence gives large positive deltas, while abstain calibration is mixed with a 0.0044 overall delta.

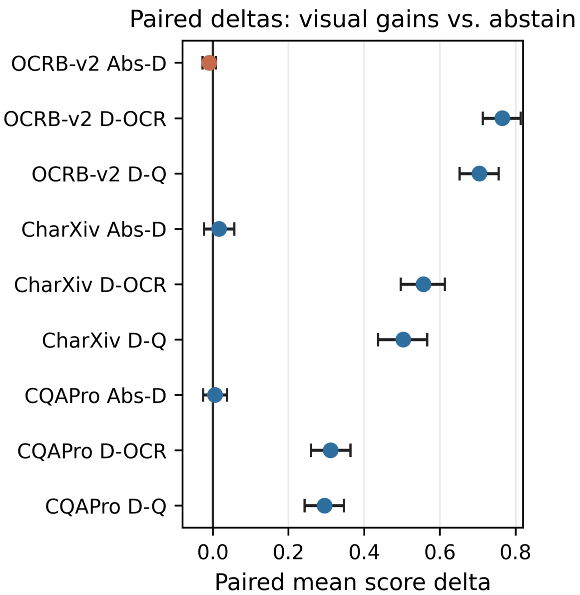

Figure 3: Paired deltas show large visual-evidence gains and near-zero abstain effects.

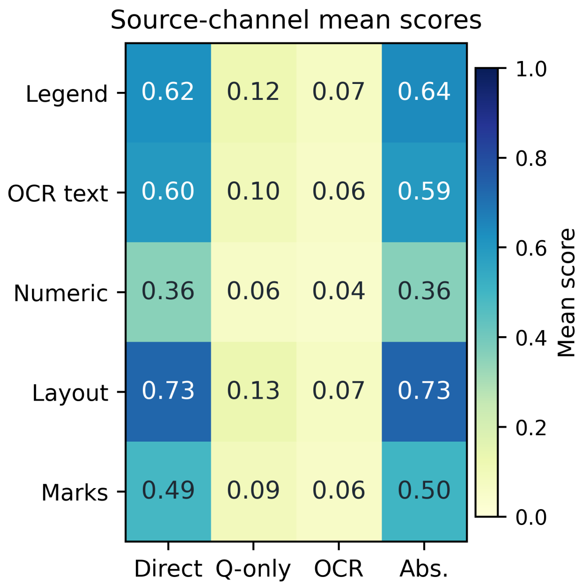

Figure 4: Source-channel heatmap shows higher image-conditioned scores across OCR text, marks, legend, numeric, and layout slices; Direct=visual, Q-only=question only, OCR=OCR text only, Abs.=abstain.

the two large visual-evidence contrasts from the near-zero abstain contrast, so the surrounding analysis can treat source-channel gains and prompt-calibration behavior as different phenomena.

Figure 4 summarizes source-channel score profiles. The slices are descriptive: they group tasks by the evidence type that the source family makes salient. OCR-visible text and text-layout reasoning remain difficult for text-only controls because the transcript is often incomplete or lacks spatial grounding. Visual mark/value extraction and legend binding also require more than a question prior. These patterns are strongest for OCR-text-only controls, where several field-grounded failure rates exceed 90%.

The source-channel view gives two warnings for future benchmark expansion. OCR-visible text is not equivalent to text-only solvability: in the automatic OCR-visible slice, OCR-only reaches 0.0582, below Q-only at 0.0965 and far below image-conditioned rows. Abstain calibration tracks image-conditioned performance closely across slices, but it does not repair the text-control gap. The shortfall summary in the canonical artifacts makes the same point operationally: OCR-only mean scores are 0.0446, 0.0667, and 0.0718 on the representative ChartQAPro, CharXiv, and OCRBench-v2 slices, with corresponding shortfall rates of 97.0%, 93.3%, and 93.0%.

These diagnostics bound interpretation. ChartQAPro remains challenging even for direct visual prompting, with a 70.0% shortfall in the combined visual/text/numeric slice, so the paper avoids "solved" language. OCRBench-v2 shows the largest separation: direct visual shortfall is 18.7%, while OCR-text-only shortfall is 93.0%. CharXiv supports a realistic chart-understanding slice and a 30/30 duplicate-grader consistency audit, but it remains PairScorer-assisted rather than a fully automatic metric claim.

Claim scope. The supported claims are deliberately narrow: matrix coverage, direct-visual improvement over question-only, direct-visual improvement over OCR-text-only, mixed abstain behavior, source-channel failure patterns, and the CharXiv scoring-scope caveat. The unexecuted comparisons are also explicit: broader open-model coverage beyond LLaVA-1.5-7B, broader chart-specialized coverage beyond ChartInstruct-FlanT5-XL, derendering baselines, fully automatic

CharXiv scoring, and OCRBench core-image evidence. Future matrices with stronger open, hosted, or chart-specialized systems should keep question-only, OCR-only, and scoring-scope rows before compressing results into a single leaderboard.

The same rule applies to reuse of the package. A new model row is comparable only when it preserves the task split, answer budget, scorer label, and source-channel fields used here. A new benchmark family should likewise report whether its score is automatic, assisted, or manually audited. These bookkeeping details are not administrative overhead; they are what prevent a chart/OCR benchmark from mixing visual grounding, OCR transcript recovery, and scorer assumptions into one opaque number.

7 Conclusion

ChartOCR-SourceDiag shows that source-channel controls change how chart/OCR robustness should be read. In a 6,300-row executed matrix over three real source families and seven conditions, hosted visual prompting strongly outperforms question-only and OCR-text-only controls, while open-weight/chart-specialized baselines and abstain-calibrated prompting expose different limits. The benchmark package is most useful as a diagnostic scaffold: it tells future evaluations where image evidence, OCR text, scoring scope, and overclaim boundaries should be separated before reporting a single aggregate number.

Limitations

This paper reports a scoped seven-condition matrix, not a broad comparison over open and API MLLMs. It includes one open image-conditioned baseline, one second hosted visual backend, and one chart-specialized baseline, but it does not cover a broad open-model sweep, multiple hosted model families, or chart-specialized derendering/programmatic systems. The OCR-family evidence is OCRBench-v2, not OCRBench core. CharXiv is LLM-assisted and only has duplicate-grader consistency evidence, not an independent human audit. Source-channel labels are descriptive and overlapping rather than causal interventions.

These limits affect interpretation directly. A weak LLaVA or ChartInstruct row in this matrix should be read as evidence about the executed baselines, not as a claim about all open or chart-specialized systems. Likewise, the CharXiv slice can support scorer-labeled diagnostic analysis, but it cannot be merged with automatic ChartQAPro and OCRBench-v2 rows as if all sources used the same oracle. The resource is therefore strongest as a protocol for controlled comparison and weakest as a standalone leaderboard.

Ethical Considerations

The benchmark sources contain public chart, document, and visual-text examples from existing dataset releases. The paper reports diagnostic aggregate behavior and does not attempt to infer sensitive attributes about people. The main ethical risk is overclaiming model reliability from a narrow backend matrix or from LLM-assisted scoring. The study mitigates that risk by separating scoring scopes, naming missing baselines, and avoiding broad deployment or SOTA claims.

Release practice should preserve those boundaries. Reproducibility bundles should store source URLs, splits, scorer labels, aggregate tables, and sanitized scored rows, while respecting each dataset's image redistribution terms. Hosted-backend versions, answer budgets, and scorer-scope labels should be logged for comparison, but private endpoints, credentials, and local execution details should not be exposed. Future model rows should retain failed or empty outputs under the same scoring policy so that error handling does not become an unreported source of improvement.

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A Reproducibility Interface

The paper-facing regeneration interface has three checks. First, rebuild the aggregate tables and result figures from the completed scored rows. Second, verify source-family, scorer-scope, and figure-asset records against the released package metadata. Third, check claim/evidence records against the aggregate result tables. The reproducibility package reports public benchmark sources, splits, scorer labels, evaluated hosted model identifiers, decoding budgets, and aggregate artifacts; lower-level execution settings are outside the paper-facing protocol.

B Additional Artifact Inventory

The canonical result artifacts are the main results table, paired-significance table, source-channel slices, failure-taxonomy summary, claim/evidence records, scorer-scope records, figure metadata, and aggregate result records. The raw scored rows contain one record per task and condition; the public paper tables use aggregate source-family and condition summaries derived from those rows.